

**Agentic AI-Based Dynamic Tariff
Optimization for EV Charging Networks
Using Large-Scale Charging Session
Data.**

Executive Summary

- **Problem:** EV charging networks running on flat ₹15/kWh pricing face chronic inefficiencies, 61% of charging slots are underutilized while peak-hour congestion drives user drop-off and revenue loss.
- **Approach:** Built a 3-agent Agentic AI framework on 2 real-world datasets (ACN: 14,848 sessions; ST-EVCDP: 2.13M grid-time slots across 247 Shenzhen grids):
 - **Demand Prediction Agent:** forecasts station utilization using Gradient Boosting on lag + temporal + spatial features
 - **Tariff Pricing Agent:** translates forecasts into real-time dynamic tariffs (₹10.5–₹30/kWh) with surge and discount triggers
 - **Monitoring & Learning Agent:** evaluates every pricing decision against actual outcomes and tracks system improvement over time

- **Key results:**

What	Result
Demand forecast accuracy	$R^2 = 0.9944$
Revenue uplift vs ₹15/kWh baseline	+17.94%
Additional revenue (5-day test window)	₹15.91 Cr
Off-peak sessions gained	+763,853
Pricing efficiency improvement	₹61.07 → ₹72.01/unit

- **Bottom Line** A self-improving dynamic pricing engine can generate ~18% more revenue with zero new infrastructure, while redistributing demand away from congested slots —benefiting operators, users, and the grid simultaneously.

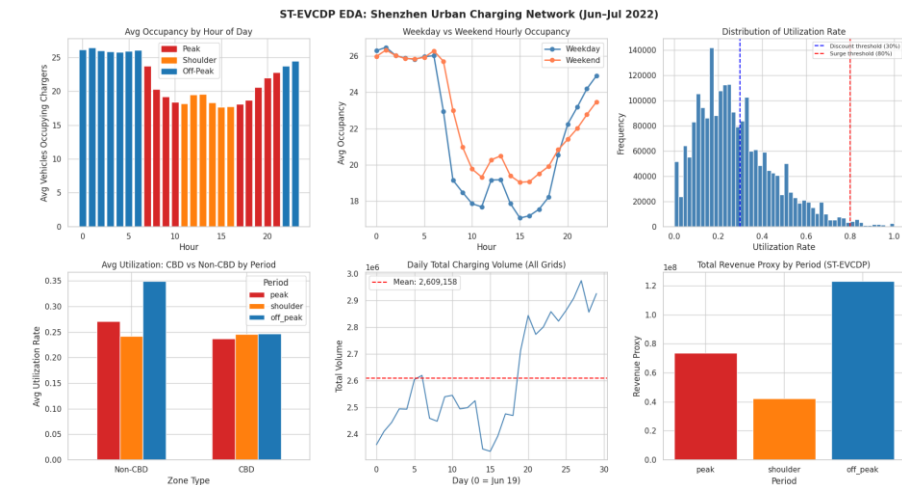
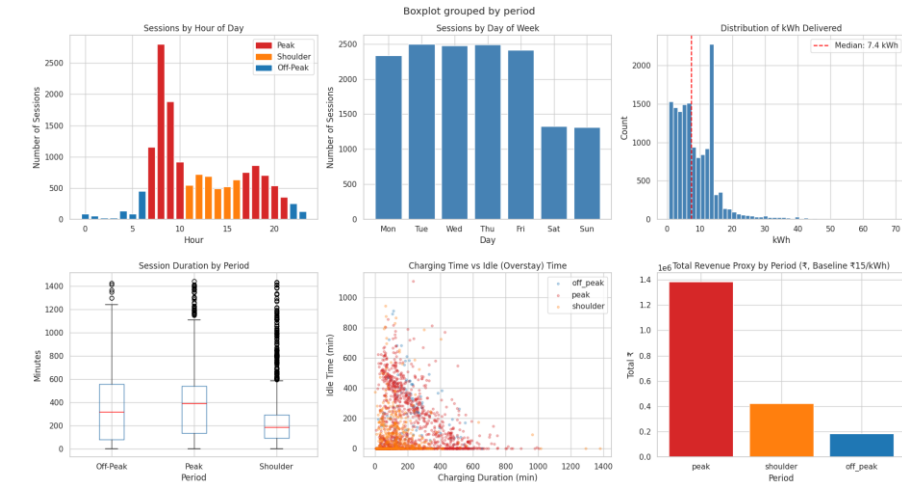
Data Landscape & Preprocessing

- Two datasets: ACN (14,848 sessions, Caltech/JPL, 2018) + ST-EVCDP (2.13M slots, 247 grids, Shenzhen, Jun–Jul 2022)
- ACN: dropped 6 fully-null columns, 1,305 missing-core rows, 151 anomalous sessions
- ST-EVCDP: melted wide→long, merged occupancy/ volume/ duration/ price, zero nulls
- Features engineered: utilization rate, idle time, peak/shoulder/off-peak labels, congestion flag (>80%), discount flag (<30%), revenue proxy
- ML modelling uses ST-EVCDP exclusively — 5-min granularity and 247-grid spatial coverage are prerequisites for lag-based forecasting. ACN used for behavioral EDA and revenue baseline.

	ACN-Data	ST-EVCDP
Source	Caltech/JPL, California	Shenzhen, China
Coverage	14,848 sessions	2.13M grid-time slots
Period	Apr–Dec 2018	Jun–Jul 2022 (30 days)
Granularity	Per session	5-minute intervals
Key fields	kWh delivered, timestamps, station ID	Occupancy, volume, duration, price
Stations	1 site (Caltech)	247 grids, 1,706 stations
Use in project	User behavior, idle time, revenue baseline	Demand prediction, dynamic tariff optimization

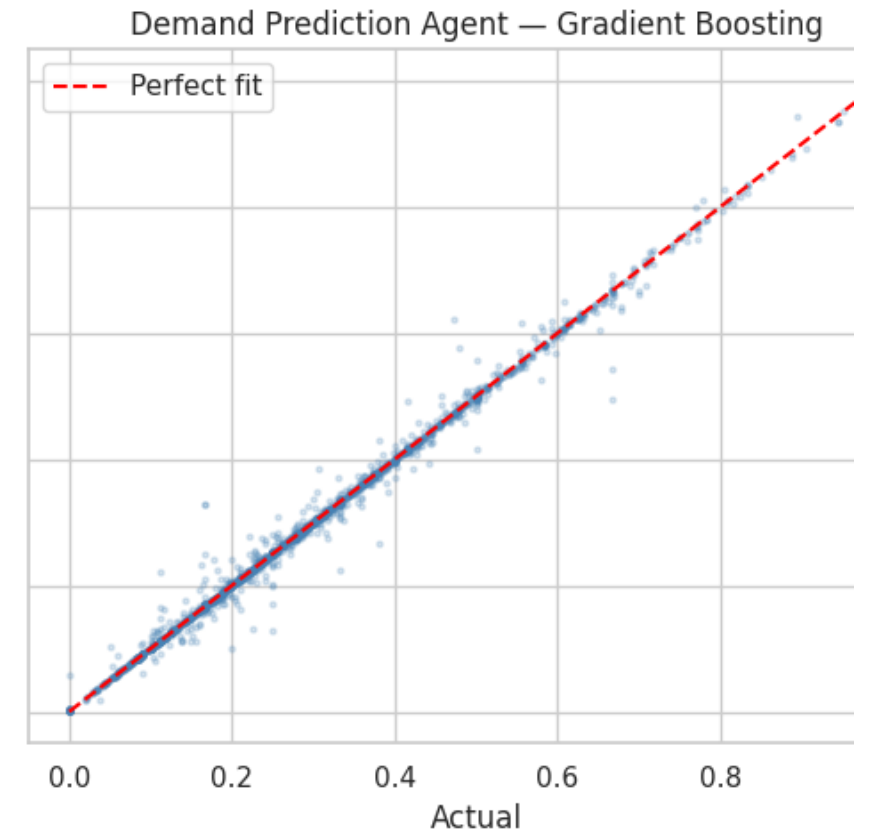
EDA: Demand Behaviour Insights

- ACN: morning spike at 8am (workplace charging), 82% weekday, avg idle overstay = 140 min (42% of session), peak drives 69.5% of revenue
- ST-EVCDP: off-peak (32%) > peak (26%) utilization Shenzhen charges at night; only 1% surge slots vs 61% discount slots; volume trending upward over 30 days
- Fleet vs. public segmentation not possible in ACN (single Caltech workplace site); ACN behavior is consistent with fleet/corporate charging signatures highly predictable weekday morning patterns, low weekend usage (17.7%)
- **chronic underutilization is the dominant inefficiency, not congestion**



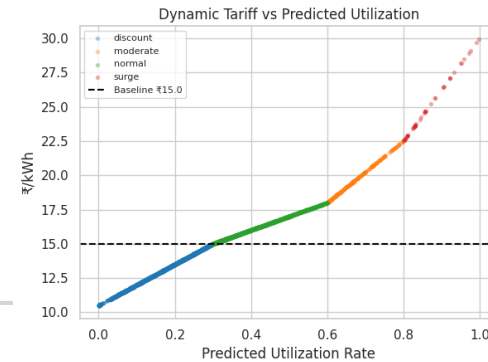
Demand Prediction Agent

- Model: Gradient Boosting (best of 3)
- RMSE: 0.0133 | MAE: 0.0051 | R^2 : 0.9944
- Temporal train/test split: last 5 days held out
- Even a naive baseline (predicting lag1 directly) achieves $R^2=0.99+$, confirming that 5-minute demand is highly autocorrelated. The GB model's marginal improvement over baseline quantifies the added value of temporal and spatial features.
- Naive lag1 baseline achieves lower MAE (0.0043 vs 0.0051) because it predicts small errors well. GB improves on RMSE penalizing large errors more heavily which matters more for pricing decisions where congestion spikes cause revenue loss.

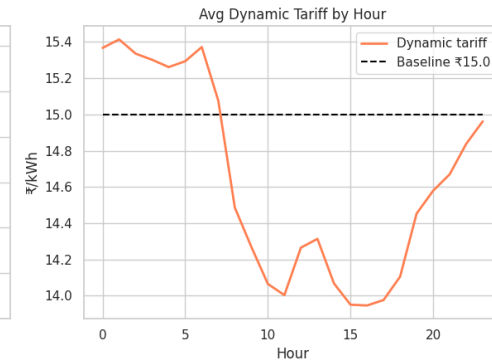
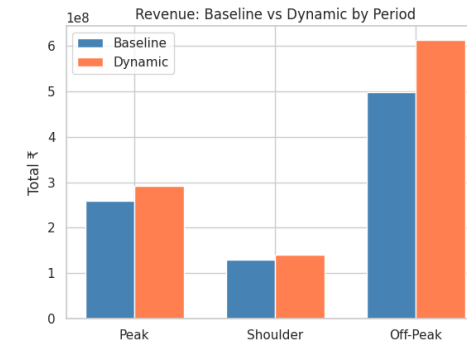


Tariff Pricing Agent

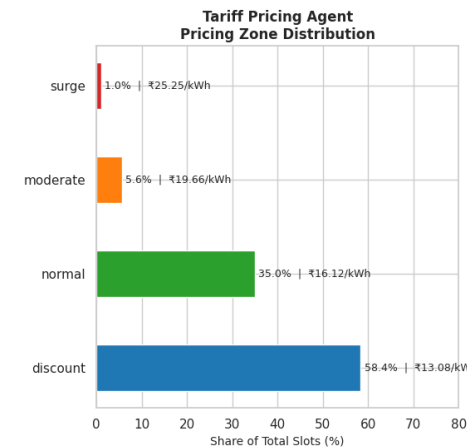
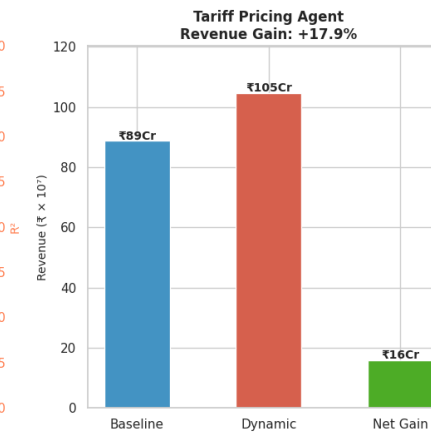
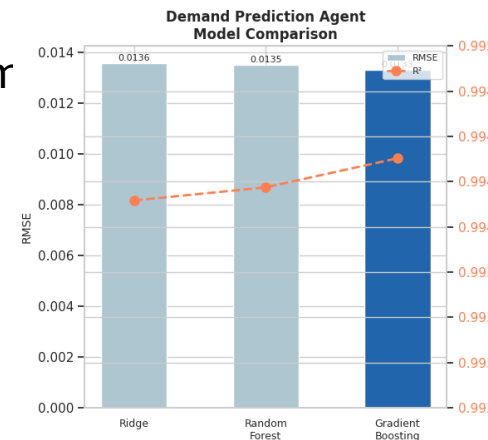
- Tiered dynamic pricing: surge (>80% util) → up to ₹30/kWh | discount (<30%) → ₹13.08/kWh | baseline ₹15/kWh
- Revenue gain: **+17.94%** (₹89Cr → ₹105Cr over 5 days)
- 58.3% slots in discount zone, confirms room to stimulate demand
- Off-peak uplift: +27.1% sessions (+763,853) [simulated, ε = empirically estimated at -2.118 (log-log regression, ST-EVCDP)] from discount pricing



Tariff Pricing Agent : Dynamic Pricing Outcomes



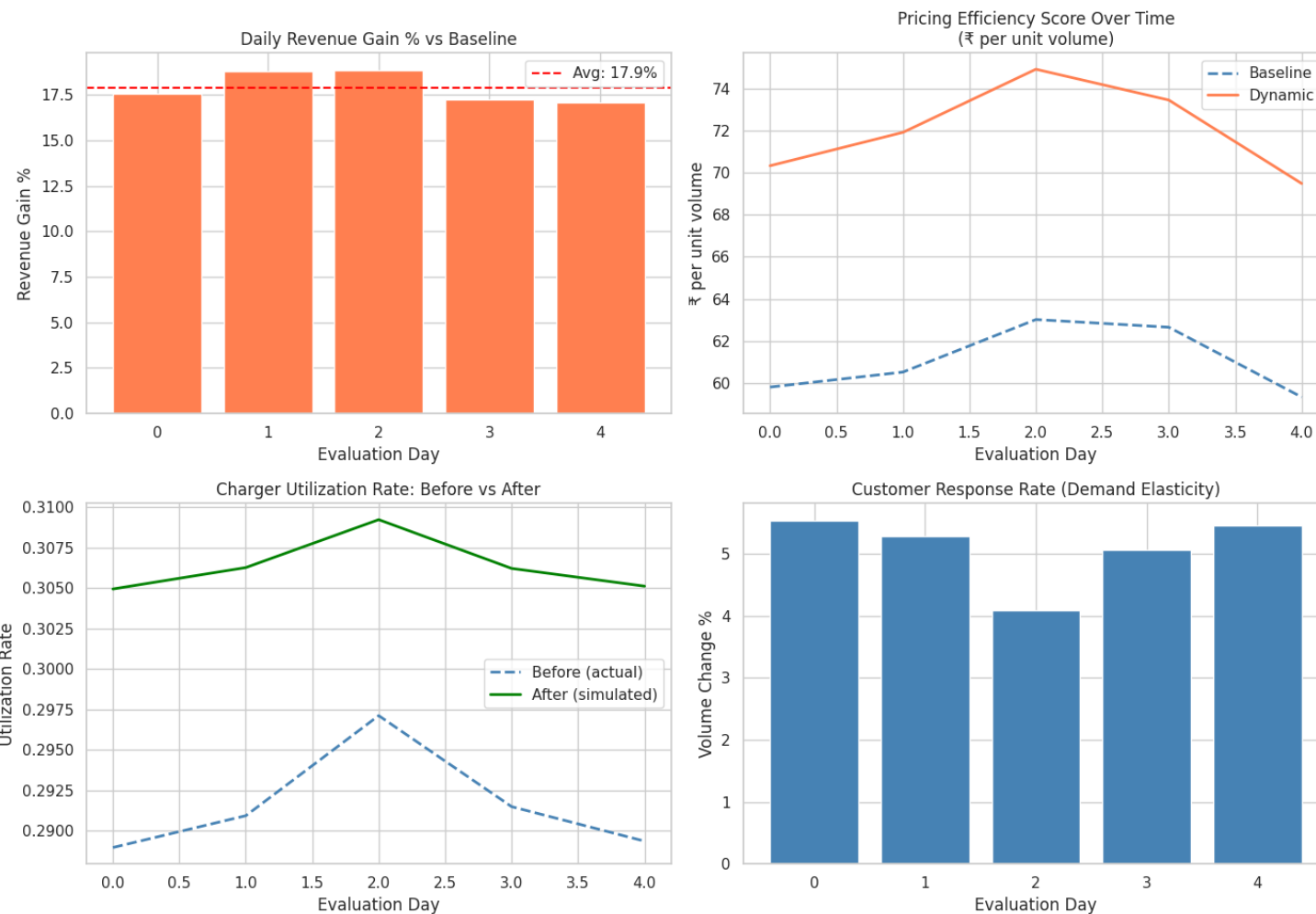
Agentic AI: Dynamic Tariff Optimization: Final Summary



Monitoring & Learning Agent

- Daily feedback loop tracked over 5 evaluation episodes
- Revenue gain stable: 17.1%–18.8% daily, system is consistent
- Pricing efficiency: ₹61.07 → ₹72.01 per unit volume (+17.93%)
- Customer response rate: +5.07% avg volume shift from tariff changes
- Wait time proxy reduced at surge slots

Monitoring & Learning Agent : Feedback Loop Performance



Business, Operational & Policy Implications

- **For operators:** 17.94% more revenue projected under conservative elasticity assumptions (ε = empirically estimated at -2.118 (log-log regression, ST-EVCDP))
- **For grid:** Discount pricing redistributes 764K sessions to off-peak → reduces grid stress during peak hours
- **For users:** Surge at only 1% of slots, 99% of the time pricing is at or below baseline
- **Policy angle:** Dynamic tariff framework can be mandated by regulators as a demand-side management tool
- Revenue gain is concentrated in surge/moderate zones (+57.7% and +36.2%); discount zones generate negative revenue per unit (-5.8%) but serve demand redistribution goals, a deliberate trade-off, not a system failure.
- **Limitations:** ST-EVCDP is 30-day window only; ACN is single-site (Caltech); no real-time grid price data integrated

Appendix A : Metrics Table

COMPLETE EVALUATION METRICS SUMMARY

Agent	Metric	Value	Interpretation
Demand Prediction	RMSE	0.0133	Very low prediction error on utilization rate
Demand Prediction	MAE	0.0051	Avg absolute error of 0.5% utilization
Demand Prediction	R ² Score	0.9944	Model explains 99.4% of demand variance
Tariff Pricing	Revenue Gain %	+17.94%	₹159,101,810 additional revenue over 5 days
Tariff Pricing	Charger Utilization (Before→After)	29.2% → 37.1%	Demand redistribution improves utilization
Tariff Pricing	Off-Peak Uplift	+27.1% (+763,853 sessions)	764K additional sessions in off-peak windows
Monitoring & Learning	Avg Wait Time Reduction (min/slot)	0.0042	Queue reduction at congested stations
Monitoring & Learning	Customer Response Rate	+5.07%	Positive demand response to discount pricing
Monitoring & Learning	Pricing Efficiency Gain %	+17.93%	₹61.07 → ₹72.01 per unit volume delivered

Appendix B: Pricing Zone Breakdown

PRICING ZONE BREAKDOWN

pricing_zone	slots	slot_share_pct	avg_util	avg_tariff	rev_gain_pct
discount	207495	58.337551	0.171782	13.077905	-5.817507
moderate	19839	5.577767	0.674079	19.669724	36.241472
normal	124632	35.040486	0.412731	16.126485	10.593734
surge	3714	1.044197	0.874716	25.250288	57.685935

Appendix C: Model Comparison

Model	RMSE	MAE	R ²
Ridge Regression	0.0136	0.0054	0.9941
Random Forest	0.0135	0.0049	0.9942
Gradient Boosting	0.0133	0.0051	0.9944
Naive Baseline (lag1)	0.0139	0.0043	0.9939

Appendix D: Baseline vs Gradient Boost

BASELINE vs ML MODEL

Naïve baseline (predict = lag1)

RMSE : 0.0139

MAE : 0.0043

R^2 : 0.9939

Gradient Boosting ML model

RMSE : 0.0133 (improvement: 4.23%)

MAE : 0.0051 (improvement: -19.78%)

R^2 : 0.9944 (improvement: 8.28%)